January 2024 | CSE 106

Online on Heaps (B1/B2)

Time: 35 Minutes

Problem

Bob finds **N** magical bags of candies where each bag contains a different number of candies. Bob picks up a bag, eats all the candies in it, and drops it on the ground. But as soon as he drops the bag, the number of candies in that bag increases magically! Say the bag that used to contain x candies (before eating), will contain $\lfloor \boldsymbol{x/2} \rfloor$ candies (after eating).

Thanks to the magical spell, Bob can now have a lot more candies! But he has to return home in **K** minutes. In a single minute, Bob can consume all the candies in a single bag, regardless of the number of candies in it.

Find the maximum number of candies that Bob can consume.

Input Format

- The first line contains two space-separated integers, **N** (number of bags) and **K** (total minutes).
- The second line contains N space-separated integers, the number of candies in the bags.

Output Format

- Print the maximum number of candies that Bob can consume.

Sample I/O

Input	**Output**
6 3 1 2 3 4 5 6	15
10 1 1 3 5 3 4 5 4 5 2 3	5
9 3 5 1 1 3 3 5 5 3 2	15

Explanation

For the first sample I/O: There are 6 bags and Bob has 3 minutes. The state of the bags is 1 2 3 4 5 6. For maximum consumption, Bob always chooses the bag with the highest number of candies.

<u>First Minute</u>: Bob eats all the candies from the sixth bag (**6**). The state of the bags becomes: 1 2 3 4 5 **3**

<u>Second Minute</u>: Bob eats all the candies from the fifth bag (**5**). The state of the bags becomes: 1 2 3 4 **2** 3

<u>Third Minute</u>: Bob eats all the candies from the fourth bag (**4**). The state of the bags becomes: 1 2 3 **2** 2 3

Maximum candies consumed by Bob: 6+5+4= 15

Instruction

- You need to implement the solution using only heap data structure and heap related operations.
- You might use the header file `MaxHeap.h` that you implemented in the home assignment.